# MA1522 Linear Algebra for Computing

## Lesson 21: Revision of Chapters 1 and 2

# Announcements

- Revision lessons:
  - today, Fri 6 Nov: Chapters 1 & 2,
  - Tue 10 Nov: Chapters 3 & 4,
  - Fri 13 Nov: Chapters 5–6.

- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

# Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson21` (e.g. `A1234567Z_lesson21.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 21**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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# Question 1

Let $a$ and $b$ be real numbers, and consider the linear system

$$\begin{cases} x - 3y + 4z = 2 \\ 2x - 5y + 3z = 5 \\ -3x + 8y + az = b. \end{cases}$$

Find the conditions on $a$ and $b$ such that the system

1. has no solution;
2. has a unique solution;
3. has infinitely many solutions. In this case, write down the general solution.

# Question 1 — solution: the three cases

Reduce the augmented matrix to a row-echelon form: $R_2 - 2R_1$, $R_3 + 3R_1$; then $R_3 + R_2$:

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 2 & -5 & 3 & 5 \\ -3 & 8 & a & b \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & -1 & a+12 & b+6 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & a+7 & b+7 \end{array}\right).$$

1. No solution: the last column is a pivot column, so $a + 7 = 0$ and $b + 7 \neq 0$, that is, $a = -7$ and $b \neq -7$.

2. Unique solution: every column of the coefficient matrix is a pivot column, so $a + 7 \neq 0$, that is, $a \neq -7$ ($b$ arbitrary).

3. Infinitely many solutions: the last row is zero, and column 3 is not a pivot column, so $a = -7$ and $b = -7$.

# Question 1 — solution: the general solution

3. Let $a = b = -7$. The row-echelon form is

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Column 3 is not a pivot column: let $z = s$. Back-substitute: $y = 1 + 5s$, and $x = 2 + 3y - 4z = 2 + 3 + 15s - 4s = 5 + 11s$.

$$x = 5 + 11s, \qquad y = 1 + 5s, \qquad z = s, \qquad s \in \mathbb{R}.$$

Check the third equation:
$-3(5 + 11s) + 8(1 + 5s) - 7s = -15 + 8 + (-33 + 40 - 7)s = -7 = b$. ✓ □

# Question 2

Let $t$ be a real number, and let

$$A = \begin{pmatrix} 1 & -4 & 2 \\ 4 & t^2 - 3t - 14 & 13 \\ -3 & 3t + 9 & 4 \end{pmatrix}.$$

1. Find all values of $t$ for which $A$ has an LU factorization.

2. Find an LU factorization of $A$ when $t = 1$.

# Question 2 — solution: part 1

1. Reduce, using only $R_i + cR_j$ with $i > j$: $R_2 - 4R_1$, $R_3 + 3R_1$:

$$A \to \begin{pmatrix} 1 & -4 & 2 \\ 0 & t^2 - 3t + 2 & 5 \\ 0 & 3t - 3 & 10 \end{pmatrix} = \begin{pmatrix} 1 & -4 & 2 \\ 0 & (t-1)(t-2) & 5 \\ 0 & 3(t-1) & 10 \end{pmatrix}.$$

- $t \neq 1, 2$: the $(2,2)$ entry is a nonzero pivot; $R_3 - \frac{3}{t-2} R_2$ gives a row-echelon form. LU factorization exists.
- $t = 1$: rows 2 and 3 are $(0, 0, 5)$ and $(0, 0, 10)$; $R_3 - 2R_2$ gives a row-echelon form. LU factorization exists.
- $t = 2$: rows 2 and 3 are $(0, 0, 5)$ and $(0, 3, 10)$; a row-echelon form needs $R_2 \leftrightarrow R_3$. No LU factorization: if $A = LU$, column 1 forces $\ell_{21} = 4$, $\ell_{31} = -3$, so row 2 of $U$ is $(0, 0, 5)$ and row 3 of $U$ is $(0, 3, 10 - 5\ell_{32})$, which is not a row-echelon form.

So $A$ has an LU factorization if and only if $t \neq 2$.

# Question 2 — solution: part 2

2. Let $t = 1$. Reduce by $R_2 - 4R_1$, $R_3 + 3R_1$; then $R_3 - 2R_2$:

$$\begin{pmatrix} 1 & -4 & 2 \\ 4 & -16 & 13 \\ -3 & 12 & 4 \end{pmatrix} \to \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 10 \end{pmatrix} \to \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix} = U.$$

The entries of $L$ are the negatives of the multipliers $-4$, $3$, $-2$:

$$A = LU = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check row 3 of $LU$: $-3(1, -4, 2) + 2(0, 0, 5) = (-3, 12, 4)$. ✓ □

# Question 3

Let $a, b, c$ be real numbers, and let

$$V = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}.$$

1. Show that $\det(V) = (b - a)(c - a)(c - b)$.

2. Let $y_1, y_2, y_3$ be real numbers. Show that if $a, b, c$ are distinct, then there is exactly one polynomial $p(x) = u + vx + wx^2$ with

$$p(a) = y_1, \qquad p(b) = y_2, \qquad p(c) = y_3.$$

# Question 3 — solution: parts 1 and 2

1. $R_2 - R_1$ and $R_3 - R_1$ do not change the determinant; then expand along column 1:

$$\det(V) = \det \begin{pmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{pmatrix} = (b-a)(c^2-a^2) - (c-a)(b^2-a^2).$$

Take out the factors $b-a$ and $c-a$:

$$\det(V) = (b-a)(c-a)\big[(c+a) - (b+a)\big] = (b-a)(c-a)(c-b).$$

2. The three conditions are a linear system in the unknowns $u, v, w$:

$$u + av + a^2 w = y_1, \quad u + bv + b^2 w = y_2, \quad u + cv + c^2 w = y_3,$$

that is, $V(u, v, w)^T = (y_1, y_2, y_3)^T$. If $a, b, c$ are distinct, then $\det(V) \neq 0$ by part 1, so $V$ is invertible and the system has exactly one solution $(u, v, w)^T = V^{-1}(y_1, y_2, y_3)^T$. □

# Challenge 1: Bruhat decomposition for $2 \times 2$ matrices

Let $\mathcal{B}$ be the set of invertible upper triangular $2 \times 2$ matrices, and let $W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be invertible. Show that
   - if $c = 0$, then $A \in \mathcal{B}$;
   - if $c \neq 0$, then $A = b_1 W b_2$ for some $b_1, b_2 \in \mathcal{B}$.

   (Hint: $b_1$ corresponds to a certain row operation, and $b_2$ to certain column operations.)

2. Show that $b_1 W b_2$ is never upper triangular, for any $b_1, b_2 \in \mathcal{B}$.

So every invertible $2 \times 2$ matrix lies in exactly one of $\mathcal{B}$ and $\mathcal{B}W\mathcal{B} = \{\, b_1 W b_2 : b_1, b_2 \in \mathcal{B} \,\}$.

# Challenge 1 — solution

1. If $c = 0$, then $A$ is upper triangular and $\det(A) = ad \neq 0$, so $A \in \mathcal{B}$. If $c \neq 0$, clear $a$ by the row operation $R_1 - \frac{a}{c}R_2$, then clear $d$ by the column operation $C_2 - \frac{d}{c}C_1$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \xrightarrow{R_1 - \frac{a}{c}R_2} \begin{pmatrix} 0 & b - \frac{ad}{c} \\ c & d \end{pmatrix} \xrightarrow{C_2 - \frac{d}{c}C_1} \begin{pmatrix} 0 & e \\ c & 0 \end{pmatrix} = W \begin{pmatrix} c & 0 \\ 0 & e \end{pmatrix}, \qquad e = \frac{bc - ad}{c} \neq 0.$$

Undo the two operations (inverse elementary matrices, on the left and on the right):

$$A = \underbrace{\begin{pmatrix} 1 & \frac{a}{c} \\ 0 & 1 \end{pmatrix}}_{b_1} W \underbrace{\begin{pmatrix} c & 0 \\ 0 & e \end{pmatrix} \begin{pmatrix} 1 & \frac{d}{c} \\ 0 & 1 \end{pmatrix}}_{b_2} = \begin{pmatrix} 1 & \frac{a}{c} \\ 0 & 1 \end{pmatrix} W \begin{pmatrix} c & d \\ 0 & e \end{pmatrix}.$$

2. Let $b_1 = \left(\begin{smallmatrix} t & u \\ 0 & s \end{smallmatrix}\right)$, $b_2 = \left(\begin{smallmatrix} p & q \\ 0 & r \end{smallmatrix}\right) \in \mathcal{B}$, so $s, p \neq 0$. Then $Wb_2 = \left(\begin{smallmatrix} 0 & r \\ p & q \end{smallmatrix}\right)$, and the $(2,1)$ entry of $b_1 W b_2$ is $0 \cdot 0 + s \cdot p = sp \neq 0$: $b_1 W b_2$ is not upper triangular. □

# Challenge 2: Bruhat decomposition for $3 \times 3$ matrices

Let $\mathcal{B}$ be the set of invertible upper triangular $3 \times 3$ matrices.

Show that every invertible $3 \times 3$ matrix $A$ can be written as

$$A = b_1 P b_2, \qquad b_1, b_2 \in \mathcal{B},$$

where $P$ is one of the six $3 \times 3$ permutation matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

and that $P$ is uniquely determined by $A$.

# Next lesson

Next lesson (**Tue 10 Nov**), we continue the revision with

1. Chapter 3: Euclidean Vector Spaces, and

2. Chapter 4: Subspaces Associated to a Matrix.

Please bring the questions you would like to go through.